



RAGSPRO DIGITAL SOLUTIONS

Grazia Stones — AI Smart Estimator

Complete App Documentation & Feature Flow

Project Ref: GRAZIA-AI-AR-2026-08

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1. Progress Summary — This Sprint (Aug 23–28)

UI/UX + App Shell Complete

Backend Integration Next

Where we are: Every screen in the app is built and running live on a physical iPhone (tested on iPhone 13 Pro this week) — full navigation, native feel, and interactions are working end-to-end. Backend wiring for Live AR, AI Visualization, and the Admin Panel is the active work for next week.

✓ Done This Sprint

- ✓ All app screens built — onboarding, login/register, home, catalogue, search, stone detail, Live AR, AI Studio, measurement, cart, checkout, orders, wishlist, dealer locator, quotes, sample orders, profile, settings.
- ✓ Native bottom navigation, page transitions, and gestures matching iOS/Android platform conventions.
- ✓ Haptic feedback wired throughout — wall lock, button taps, tab switches, and key actions on Live AR.
- ✓ Native iOS back-swipe and Android back-button handling on every screen.
- ✓ App installed and running live on a physical iPhone 13 Pro — verified cold start, navigation, and screen transitions on real hardware, not just the simulator.
- ✓ Core native AR foundation in place on iOS (ARKit) — world tracking and wall detection running on-device.

→ Next Week

- Finish the Admin Panel — product, order, and quote management from one dashboard.
- Wire the backend for Live AR — full wall-lock-to-checkout data flow connected to the live database.
- Wire the backend for AI Visualization (room photo mode) end-to-end.
- You add the real product catalogue (tiles/stones, prices, images) — the app will pull it live once added.
- Set up the 3D wall visualizer (elevation + perspective view).

2. Project Overview

Grazia Stones AI Smart Estimator is a mobile + web application that lets a customer point their phone at a real wall, see a chosen tile/stone design projected onto it in real time using AR, measure the wall directly from the camera, get an automatic tile quantity and cost estimate, and place an order or request a quote — all inside one app. It also includes an AI-based "upload a room photo" mode for customers who don't want to use live AR, plus a catalogue, cart/checkout, dealer & lead management, and an admin panel to manage products centrally.

Tech Stack

Layer	Technology
Mobile App	Flutter (iOS + Android), Riverpod (state), GoRouter (navigation)
AR — iOS	Native ARKit (Swift) — world tracking, vertical plane detection, LiDAR/scene-depth where supported
AR — Android	Native ARCore (Kotlin) — world tracking, plane detection
AI Visualization	NVIDIA NIM (vision model) for wall/object segmentation
Backend / Database	Supabase (Postgres + Auth + Storage + Realtime)
Payments	Razorpay

3. Complete App Flow

3.1 First-Time Journey

1. **Splash** — brand animation, initializes local storage and connects to backend in the background.
2. **Onboarding** — introduces the core value: point your phone at a wall, see the design live.
3. **Login / Register / Guest** — email+password or phone OTP; guest browsing supported for the catalogue.
4. **Home** — featured collections, trending stones, quick entry into Live AR.

3.2 Core Flow — Live AR Visualization

Step	What Happens
Open Live AR	Camera permission requested contextually (only here, not at app launch). Native ARKit/ARCore session starts.
Point at wall	App detects vertical wall planes in real time and shows a subtle detection indicator.
Wall locked	Once tracked with confidence, status changes from "Searching" to "Wall detected" — with haptic feedback.
Select tile/stone	Swipe through the product carousel — material projects onto the real wall instantly, no AR restart.
Switch colour/variant	Tap a colour swatch — texture updates on the wall live, tracking is not interrupted.

Move the phone	The wall anchor stays spatially fixed — perspective changes naturally as you move.
Measure wall	Tap two points on the wall (real ARKit world anchors, not screen pixels) for width, then two more for height.
Auto tile quantity	App calculates tiles needed, wastage %, and estimated cost from the measured wall area and tile size.
Capture / Share	Save or share the visualization and the estimate.

3.3 AI Room Photo Mode (no live AR needed)

1. Upload or take a photo of the room.
2. Select tile/stone + colour + finish.
3. Backend detects the wall, segments obstacles (furniture, windows, doors), applies the material only to the wall.
4. Result preserves the room's real furniture and lighting.

3.4 Catalogue → Order Flow

1. **Collections / Search** — browse by category, filter, search.
2. **Stone Detail** — full specs, price, gallery, "Try in AR" shortcut.
3. **Cart** — add tiles/stones with quantity.
4. **Checkout** — address, Razorpay payment, order confirmation.
5. **Orders** — track order status.
6. **Sample Order / Quote Request** — request a physical sample or a formal PDF quotation instead of buying directly.

3.5 Supporting Flows

Screen	Purpose
Wishlist	Save products for later.
Dealer Locator	Find the nearest Grazia Stones dealer.
Profile	Account details, addresses, order history, help & support.
Settings	Theme, notifications, unit preferences (ft/m/in/cm).

4. Feature Status

Feature	Status	Notes
All app screens / UI	DONE	Every screen built and navigable, running live on physical iPhone.
Native feel — haptics, gestures, transitions	DONE	Verified on iPhone 13 Pro this sprint.
Native AR — iOS (ARKit)	DONE	World tracking, wall detection, texture projection running on-device.
Wall measurement (world-space)	DONE	Real ARKit anchors, not screen coordinates.
Live AR — backend data flow	IN PROGRESS	Starting next week — connecting AR flow to live product/order data.
AI Visualization — backend	IN PROGRESS	Starting next week.
Admin Panel	IN PROGRESS	Being finished this week/next.
Product catalogue (real data)	NEXT	Client to provide product list next week.
3D Wall Visualizer	NEXT	Setup planned next week.
Payments (Razorpay)	IN PROGRESS	Integrated; server-side verification being finalized.

This status is kept up to date against the actual running build — every "DONE" item above has been verified on a physical device, not just in code.

5. Timeline Reference

Phase	Deliverable
Week 1–2	Backend setup, Admin Panel, product & catalogue management

Week 3–4	Real AI Visualization + Real AR (wall detection, auto tile fit)
Week 5	Wall measurement, tile calculator, cost estimator, PDF BOQ
Week 6	Testing, Play Store + App Store submission

For full commercial terms, payment schedule, and scope, refer to the signed Service Agreement (GRAZIA-AI-AR-2026-08). This document is a technical/functional progress reference.